Zishuo Zhao 赵梓硕

zishuozhao@hkust-gz.edu.cn, wiku30.0@gmail.com

Assistant Professor, FinTech Thrust, HKUST (GZ)

Background

- **2026/08 - now** Assistant Professor, FinTech Thrust, Society Hub, HKUST (GZ)
  PI of the *Trustworthy AI Mechanism & Economics Research (T'AIMER)* Lab
- **2026/03 - 2026/07** Postdoctoral Fellow, School of Computing, National University of Singapore
  Advisor: Jiaheng Zhang
- **2025/07 - 2026/01** Research Intern, Gradient Network [Website]
  Research Topic: Incentive & protocol design of decentralized AI systems.
- **2023/09 - 2024/08** Visiting Student, Institute for Data Systems and Society (IDSS), Massachusetts Institute of Technology
  Research Topic: Data-Driven Dynamic Mechanism Design
  Advisor: David Simchi-Levi
- **2021/01 - 2026/01** PhD, Department of Industrial & Enterprise Systems Engineering, University of Illinois Urbana-Champaign
  Research Area: Operations Research
  Advisor: Yuan Zhou
  Thesis Title: *Incentive Design for Digital Economy and AI Platforms* [Link]
- **2020/08 - 2021/01** Research Assistant, Haihua Institute of Frontier Information Technology
  Research Area: Network Optimization
  Advisor: Longbo Huang
- **2015/08/19 - 2020/07/24** Undergraduate Student, Institute for Interdisciplinary Information Sciences (Yao Class), Tsinghua University

Research Interest

I am an Assistant Professor at HKUST (GZ), leading the *Trustworthy AI Mechanism & Economics Research (T'AIMER)* Lab [Link], focusing on interdisciplinary research in blockchain, decentralized trustworthy AI, AI economics, AI security/safety, and other related fields. My general research interest lies in the incentive design for digital economy and AI platforms, with applications in both centralized and decentralized platforms.

I have a special interest in the mechanism design for blockchain and decentralized economics, with a (starry-eyed) dream [i*] on the exploration of the design and theoretical boundaries of **decentralized trustworthy AI** platforms, including (particularly) the current concern and initiative of **AI safety** for the upcoming AGI era.

Particularly, I am recently interested in the exploration into a novel paradigm of *incentive security* on blockchain and AI platforms, which aims to combine game-theoretic and systems/cryptographic methodologies to prevent dishonest behavior of *untrusted but rational* parties.

Selected Papers

Bayesian Mechanism Design for Blockchain Transaction Fee Allocation [Link]

Xi Chen*, David Simchi-Levi*, Zishuo Zhao*, Yuan Zhou*. (alphabetical order)

Operations Research, 73(4), 1944-1964, 2025.

Best Paper Award, *NeurIPS Workshop on Decentralization and Trustworthy Machine Learning in Web3 (NeurIPS DMLW'22).*

Crypto Economics Security Conference (CESC 2022).

Invited to OR Talk by 运筹OR帷幄 (OR China) in 2024.

Invited to *INFORMS Annual Meeting 2022.*

It Takes Two: A Peer-Prediction Solution for Blockchain Verifier's Dilemma [i*] [Link]

Zishuo Zhao, Xi Chen, Yuan Zhou.

Major revision in *Operations Research.* (2026)

Invited to the *BDePIN-Energy'26* workshop in the *8th Conference on Blockchain Research & Applications for Innovative Networks and Services (BRAINS 2026).*

Invited to OR Talk by 运筹OR帷幄 (OR China) in 2024.

Proof-of-Learning with Incentive Security [i*] [Link]

Zishuo Zhao, Zhixuan Fang, Xuechao Wang, Xi Chen, Hongxu Su, Haibo Xiao, Yuan Zhou.

In Submission (2026).

Invited to *INFORMS Annual Meeting 2025.*

Invited to *INFORMS Conference on Security (IConS'24).*

ACM EC Workshop on Foundation Models and Game Theory (ACM EC FMGT'24).

A preliminary version presented on *INFORMS Annual Meeting 2023.*

Other Papers

Multi-agent Adaptive Mechanism Design [Link]

Qiushi Han*, David Simchi-Levi*, Renfei Tan*, Zishuo Zhao*. (alphabetical order)

The 22nd Conference on Web and Internet Economics (WINE 2026).

Best Paper Award, Statistics & Information session, 31st Annual LIDS Student Conference at MIT. (2026)

VeriLLM: A Lightweight Framework for Publicly Verifiable Decentralized Inference [Link]

Ke Wang*, Zishuo Zhao*, Xinyuan Song, Zelin Li, Libin Xia, Chris Tong, Bill Shi, Wenjie Qu, Eric Yang, Lynn Ai. (equal contribution)

Personalized Pricing with Group Fairness Constraint [Link]

Xin Chen*, Zexing Xu*, Zishuo Zhao*, Yuan Zhou*. (alphabetical order)

ACM Conference on Fairness, Accountability, and Transparency (ACM FAccT 2023)

Dynamic Car Dispatching and Pricing: Revenue and Fairness for Ridesharing Platforms [Link]

Zishuo Zhao, Xi Chen, Xuefeng Zhang, Yuan Zhou

International Joint Conference on Artificial Intelligence (IJCAI 2022), **Long Oral (3.75%)**.

Invited to *INFORMS Annual Meeting 2021.*

ClusterSLAM: A SLAM Backend for Simultaneous Rigid Body Clustering and Motion Estimation [Link]

Jiahui Huang, Sheng Yang, Zishuo Zhao, Yukun Lai, Shi-Min Hu.

Computational Visual Media (CVM), Volume 7, pages 87–101 (2021).

International Conference on Computer Vision (ICCV 2019).

Awards

Academic

- **Best Paper Award**, Statistics & Information session, 31st Annual LIDS Student Conference at MIT, 2026
- **Best Paper Award**, NeurIPS Workshop on Decentralization and Trustworthy Machine Learning in Web3, 2022

- Crypto Economics Security Conference (CESC) Travel Award, 2022
- UIUC Graduate College Conference Presentation Award, 2021
- Xuetang Scholarship in Tsinghua University, 2016-2020
- 12th place in 2nd THUCTF Cybersecurity Contest in Tsinghua University, 2020
- 12th place in 24th Artificial Intelligence Programming Contest in Tsinghua University, 2020
- 15th place in 20th Artificial Intelligence Programming Contest in Tsinghua University, 2016.
- Second Prize in Chinese Mathematical Olympiad (CMO), 2014
- First Prize in National Olympiad in Informatics in Provinces (NOIP), 2013

Arts

- Finalist Award in the "一盏茶时 (*IIIS Tea Time*)" photography exhibition for the 12th Anniversary of IIIS, Tsinghua University, 2023.
- Scholarship for Arts Excellence in Tsinghua University, 2018.
- Third Prize in the Art Festival of No.1 Middle School affiliated to CCNU, for piano performance *Hatsukoi*, 2013.

Invited Talks & Academic Services

- Served as the committee chair in the postgraduate qualification examination at HKUST(GZ) in Oct 2026.
- Invited by Yau Mathematical Sciences Center, Tsinghua University for the talk "Decentralized AI: Economics of Trust" in Feb 2026.
- Invited to MIT DSL Seminar for the talk "Decentralized AI: Economics of Trust" in Oct 2025.
- Invited to OR Talk by 运筹OR帷幄 (OR China) for two talks in 2024.
- Facilitator in section Revenue & Pricing, INFORMS Annual Meeting, 2021.
- Invited to the Alumni Forum for the 10th Anniversary of IIIS, Tsinghua University in 2021.

Conference & Journal Reviewing

- Journal Reviewing: Operations Research (2025), Digital Finance (2026)
- Conference Reviewing: ITCS(2027), ICML (2026), NeurIPS (2026), Web3 Scholars Conference (2026), ESA (2024)

Teaching

- Guest lecture on decentralized AI, IE598 Foundations of Modern Machine Learning, UIUC, Fall 2025.
- Guest lecture on blockchain transaction fee mechanism, IE524 Optimization in Finance, UIUC, Fall 2025.
- Teaching Assistant, Causal & Statistical Inference, Tsinghua University, Fall 2018.
- Discussion Session, Calculus, Tsinghua University, Fall 2015.

Advising

PhD students

- Zhiyang Ding (BSc in Economics @ Peking University, 2026-now)
- Fangdu Ren (MSc in Optimization @ Université Paris-Saclay, 2026-now)
- Hongxu Su (Co-advising with Xuechao Wang)

During my PhD @ UIUC

- Renfei Tan (PhD student at MIT)
- Qiushi Han (Undergraduate at UIUC. Next step: MIT ORC PhD)
- Haibo Xiao (Undergraduate at Tsinghua University. Next step: Tsinghua PhD)

Miscellaneous

Languages

- Mandarin Chinese (native)
- English (fluent)
- French (basic)
- A few other languages on very basic levels (with amateur interest)

Coding Skills

- Mainly using C++, MATLAB and Mathematica, also with command of Python, Java, PHP.
- With some knowledge in ROS, Verilog HDL and assembly language.

Extracurricular Activities

- I have an amateur interest in Capture-The-Flag (CTF) cybersecurity competitions, and developed two Reverse challenges in the TQLCTF 2022.
- I have a wide scope of hobbies in arts and aerobic sports, especially in vocal music, photography, piano, calligraphy, table tennis and orienteering.

Selected Non-Academic Work

- During the COVID-19 outbreak, I made a Magic Tower-style indie game entitled *Expecto Patronum* in 2020, featuring a fantasy theme of a young researcher's struggles with the concern of AI safety, with 10 endings depending on the player's adventure and choices. [Link]
- I wrote my recap of 2022 in Classical Chinese, entitled 《晨星赋》, describing the second year of my PhD journey. [Link]
- The dedication of my PhD thesis was handwritten by myself on Aug 16, 2025, but I unfortunately lost the original copy when I graduated from UIUC.

(Updated on Oct 21, 2026)